

Project Summary

Naval ships are confined, high-density environments in which infectious diseases can spread before crews have accurate exposure data. We created BRIGID (Behavioral Real-time Interaction Graphing for Infectious Diseases), a hybrid UWB/BLE wearable monitoring and analytics device that tracks sailor proximity in real time and utilizes machine learning to detect high-risk exposures. BRIGID enables crews to make more targeted quarantine decisions.

Problem & Motivation

COVID-19, influenza, and norovirus epidemics have a negative impact on crew health, effecting mission readiness. Existing systems rely on symptoms, manual reporting, or delayed contact tracing, which makes interventions reactive rather than preventative. Our goal is to design a wireless system that improves on BLE-only monitoring by measuring proximity (6 ft) within ± 0.1 ft of error and provides early detection of high-risk contacts.

Experimental Setup/Device

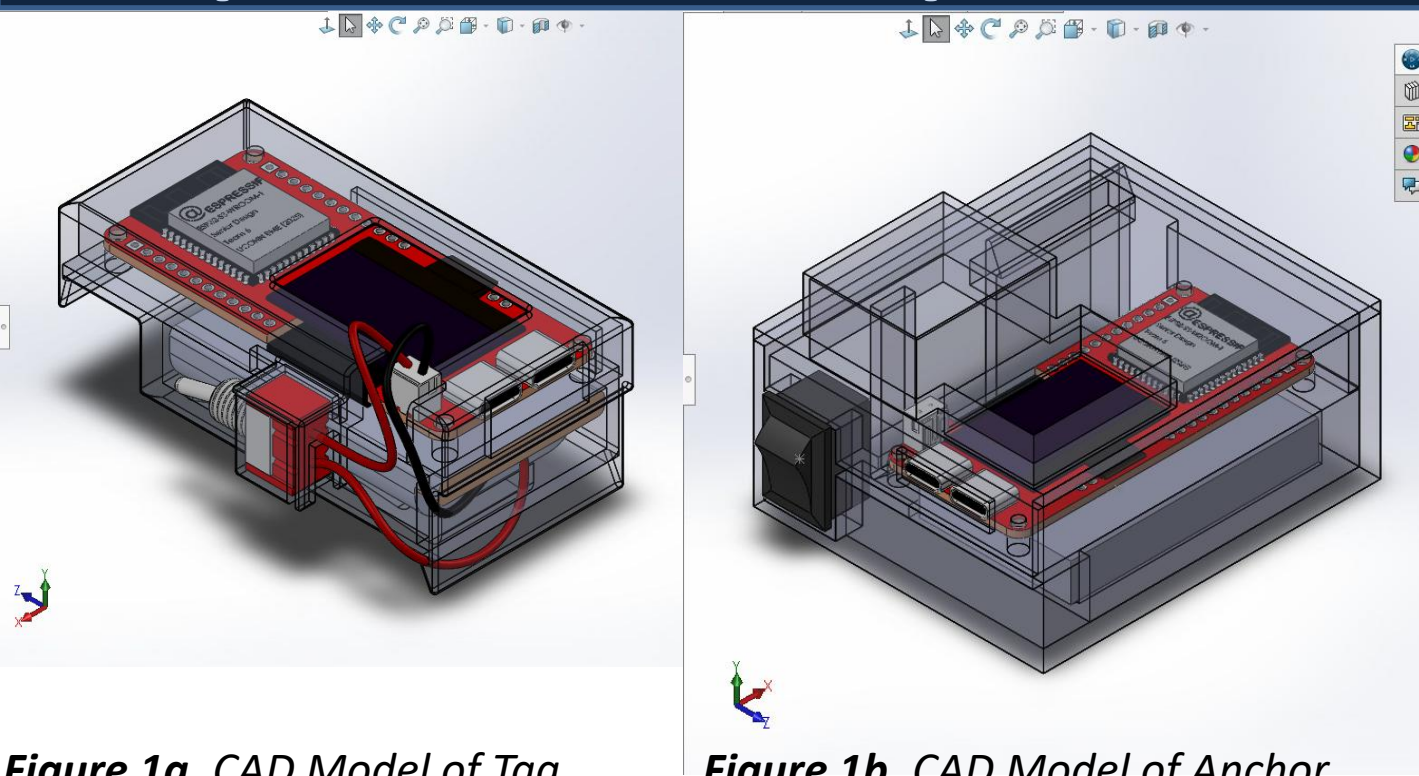


Figure 1a. CAD Model of Tag Hardware

Figure 1b. CAD Model of Anchor Hardware

- Wearable tags (Fig. 1a) employed with MaUWB ESP32S3 modules to measure the distance to fixed anchors via time-of-flight ranging.
- Fixed anchors (Fig. 1b) housed with relatively the same hardware as tags were placed at known room locations as reference points for real-time location tracking.
- A XIAO ESP32-C6 listened to tag broadcasts and sent distance strings to the central PC via serial.
- This setup enables for wireless proximity detection and contact-data logging for exposure analysis.

Data & Input Signals

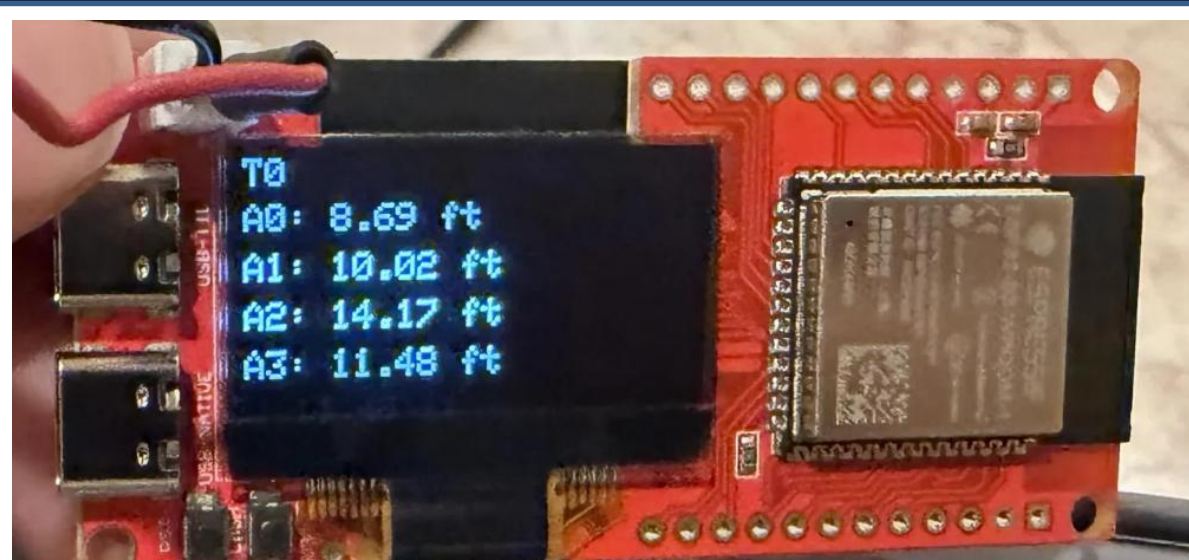


Figure 2. MaUWB ESP32S3 Displaying 4 anchor readings

- Data collected came from three wearable tags and up to four fixed anchors (Fig. 2). The Tags polled UWB ranges every 200 milliseconds and broadcast anchor-distance strings via BLE:
 - Example input: T0 | A0:3.12 | A1:8.45 | A2:--- | A3:10.20
- Preprocessing included converting ranges to feet, applying calibration, and smoothing noise. BRIGID then received the processed tag data on a local PC and converted it into timestamped (x,y) positions and contact pairs. Local CSV files were used to organize the data for exposure analysis:
- Example output: 4.01.2026 12:02:16, T1, T2, 4.4ft, 30.0s

Acknowledgements

The UConn-URI Navy STEM Coalition recognizes the critical support from the ANCHOR Program, managed by General Dynamics Electric Boat and funded by the U.S. Navy through the Maritime Industrial Base Program, which enables continued investment in workforce development and STEM education.

System Architecture

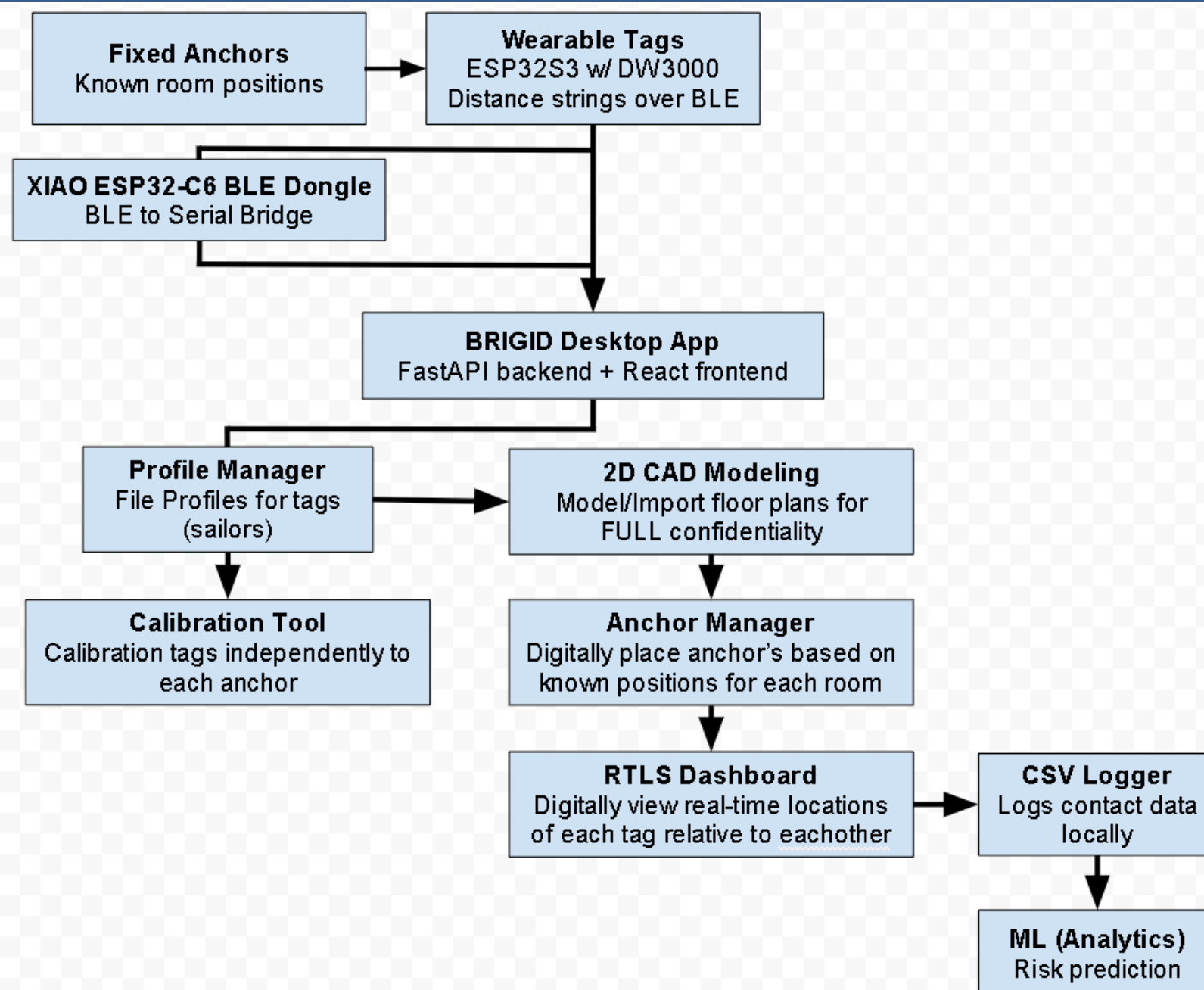


Figure 3. Block Diagram of System Architecture

Algorithm Pipeline

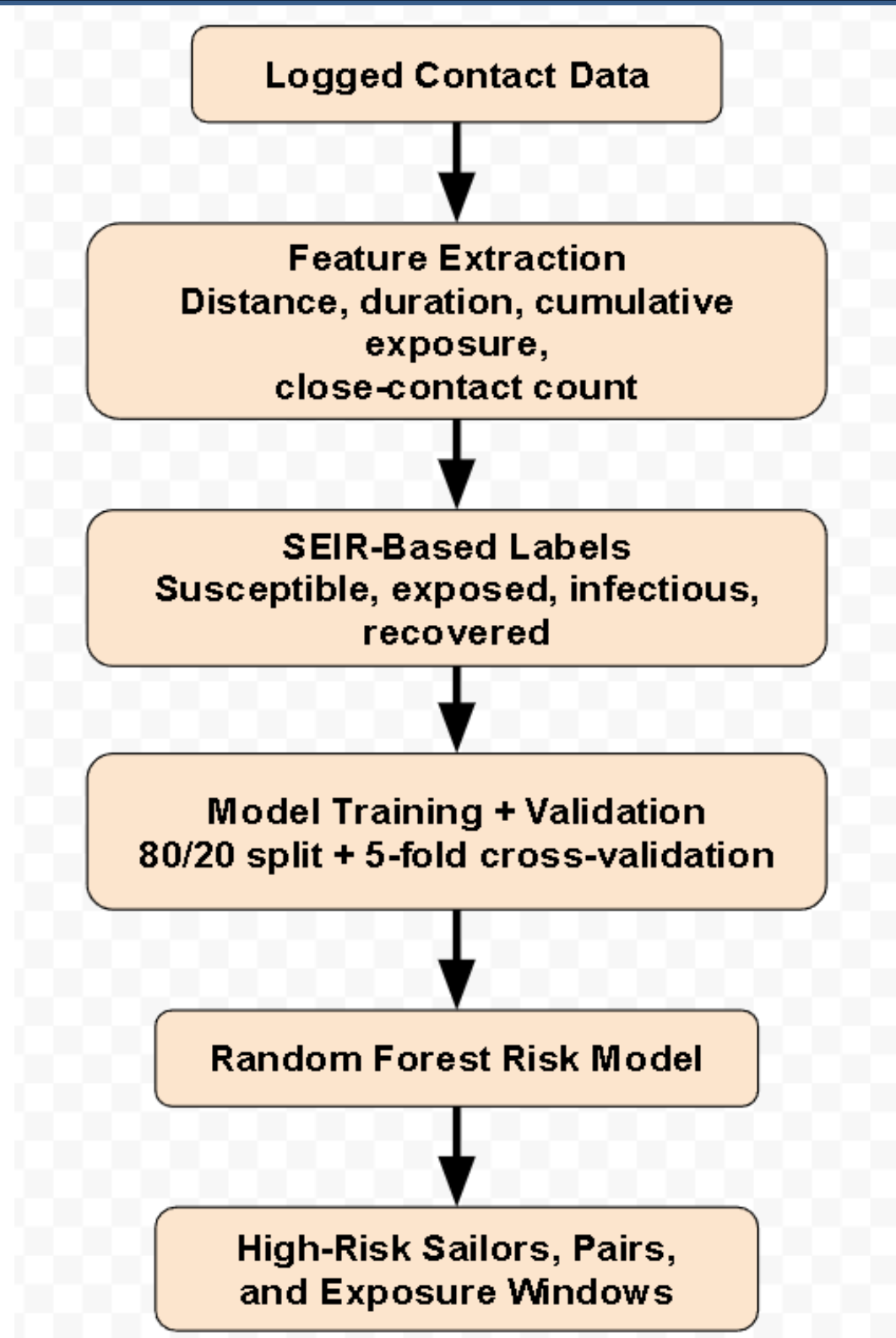


Figure 4. Flow chart of Algorithm Pipeline

BRIGID's decision logic employs the trained Random Forest model to translate exposure features into a transmission-risk probability for each sailor pair and time window (Fig. 3). The approach uses predefined thresholds to classify risk as Low < 0.35 , Elevated ≥ 0.35 , High ≥ 0.65 , or Critical ≥ 0.85 . These rated risk bands determine which sailors, tag pairs, and exposure windows are flagged for potential intervention. Assessed several models, including Random Forest, Gradient Boosting, SVM, Logistic Regression, ANN, Naive Bayes, and KNN.

Implementation Details

- Hardware Stack:** MaUWB ESP32S3 3 wearable tags + 4 fixed anchors, 7 2000-22000 mAh LiPo Batterys, 7 portable housings, and a XIAO ESP32-C6 BLE-to-serial dongle.
- Software Stack:** Arduino/C++ firmware w/ NimBLE, Python/FastAPI backend, React/TypeScript frontend, Electron desktop shell, and the scikit-learn ML runtime.
- Runtime Environment:** BRIGID operates locally on a PC, with BLE tag data bridged over serial and processed by the desktop client, eliminating the need for cloud connectivity.
- Throughput:** Tags request UWB ranges every 200 ms and provide cleaned anchor-distance strings over BLE (instant).
- Embedded Constraints:** Firmware handles calibration, smoothing, stale-data removal, BLE packet formatting, and a limited onboard display output for low-power wearables.

References

- Sun, Y., Koo, J. R., Park, M., Yi, H., Dickens, B. L., & Cook, A. R. (2023). Use of Bluetooth contact tracing technology to model COVID-19 quarantine policies in high-risk closed populations. Digital Health, 9, 20552076231178418. <https://doi.org/10.1177/20552076231178418>
- Stehlé, J., Voirin, N., Barrat, A., Cattuto, C., Colizza, V., Isella, L., Régis, C., Pinton, J.-F., Khanafer, N., Van den Broeck, W., & Vanhems, P. (2011). Simulation of an SEIR infectious disease model on the dynamic contact network of conference attendees. BMC Medicine, 9(1), Article 87. <https://doi.org/10.1186/1741-7015-9-87>

Experimental Results

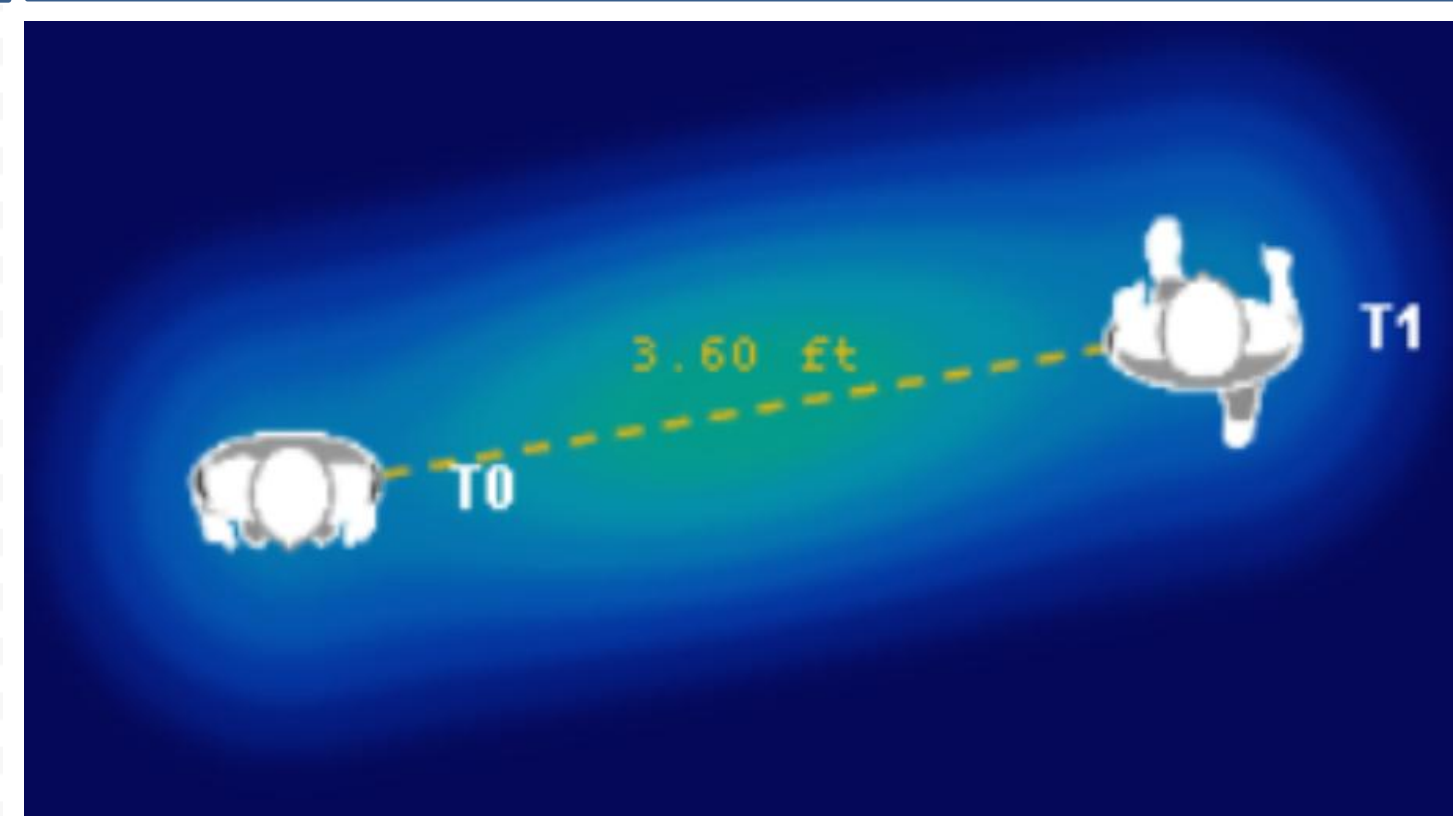


Figure 5. Image of Fully Functional RTLS, Showing 2 Sailors (tags) that are 3.60 ft from each other

Calibrated UWB ranging recorded controlled distances with an absolute error of 0.1 ft, meeting the accuracy requirements for close-contact detection. Wearable tags successfully ranged against up to four fixed anchors, streamed clear distance data over BLE, and transmitted live readings to BRIGID using the XIAO ESP32-C6 serial bridge. BRIGID transformed the receiving tag data into a 2D RTLS (Fig. 5).

Algorithm Performance

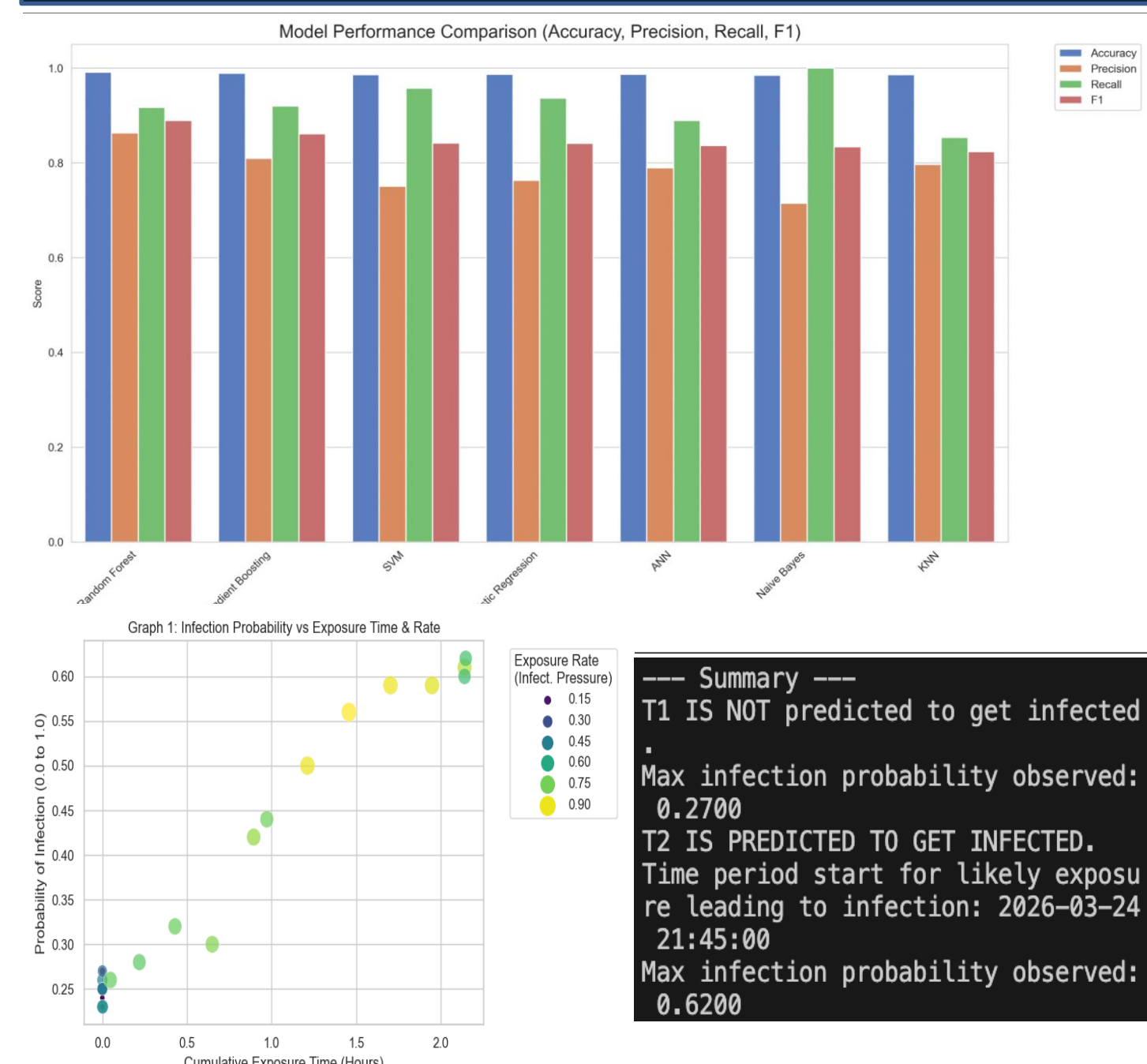


Figure 6. ML model evaluation and exposure-risk output

- 7 Machine Learning models were trained from data generated based on the SEIR model.
- Random Forest performed best, with an F1 score of 0.889, indicating the optimum balance between detecting infections and limiting false alarms.
- The model's capacity to predict high-risk exposure patterns from contact data was supported by final validation results (ROC-AUC = 0.81, recall ≈ 0.78).

Error Analysis

- UWB readings might vary due to body blocking, anchor location, multipath reflections, and signal noise.
- Absolute error fluctuates due to noise (± 0.4 ft)
- ML predictions rely on simulated SEIR labels and contact-feature quality, therefore real shipboard validation is required prior to deployment.

Limitations & Next Steps

- Redesign the wearable tag to be lighter, smaller, and more pleasant for extended shipboard usage.
- Increase power efficiency while using continuous proximity monitoring.
- Add biosensors like temperature, heart rate, and other physiological indications to help diagnose diseases earlier.
- Improve ML model by integrating larger datasets and health indications other than proximity alone.

